

Thesis: A Comparative Study of Large Language Models for Financial Sentiment Analysis and Their Predictive Potential for Short-Term Stock Price Movement

Component: Experiment 2 - Model A: FinBERT (ProsusAI/finbert)

Description: This notebook loads the prepared 1000 headline dataset and evaluates FinBERT on Experiment 2. It classifies each headline as positive or negative, maps predictions to directional signals, and computes directional accuracy, precision, recall and F1-score based on next-day stock price movement.

Select T4 GPU as runtime.

```
# Install required libraries – Experiment 3a: FinBERT

!pip install -q transformers
!pip install -q pandas scikit-learn matplotlib seaborn

print("Libraries installed successfully.")
```

Libraries installed successfully.

```
# Import required libraries – Experiment 2a: FinBERT

import pandas as pd
import numpy as np
import torch
import warnings

from transformers import pipeline
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, classification_report
)

warnings.filterwarnings("ignore")

print(f"PyTorch version : {torch.__version__}")
print(f"CUDA available : {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"GPU detected : {torch.cuda.get_device_name(0)}")
```

PyTorch version : 2.10.0+cu128
CUDA available : True
GPU detected : Tesla T4

Add the API Key from Hugging Face using "Add New Secret" in Google Colab

```
# Connect to Hugging Face and load Experiment 2 dataset

from google.colab import userdata, drive
from huggingface_hub import login

drive.mount("/content/drive")

hf_token = userdata.get("HF_TOKEN")
login(token=hf_token)

df_exp2 = pd.read_csv("/content/drive/MyDrive/Thesis_Data/exp2_dataset.csv")

print("Dataset loaded.")
print(f"Total headlines : {len(df_exp2)}")
print(f"Columns : {list(df_exp2.columns)}")
print(f"\nMovement distribution:")
print(df_exp2["movement"].value_counts())
```

Mounted at /content/drive
Dataset loaded.
Total headlines : 1000
Columns : ['ticker', 'date', 'headline', 'finbert_sentiment', 'close_price', 'next_close_1d', 'movement']

Movement distribution:
movement
1 508
0 492
Name: count, dtype: int64

```
# Load FinBERT
```

```
print("Loading FinBERT...")

finbert = pipeline(
    "text-classification",
    model="ProsusAI/finbert",
    tokenizer="ProsusAI/finbert",
    device=0 if torch.cuda.is_available() else -1
)

print("FinBERT loaded successfully.")
```

```
Loading FinBERT...
config.json: 100% 758/758 [00:00<00:00, 87.5kB/s]

pytorch_model.bin: 100% 438M/438M [00:03<00:00, 427MB/s]

Loading weights: 100% 201/201 [00:00<00:00, 646.33it/s, Materializing param=classifier.weight]
BertForSequenceClassification LOAD REPORT from: ProsusAI/finbert
Key | Status | |
-----+-----+--
bert.embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch
tokenizer_config.json: 100% 252/252 [00:00<00:00, 17.2kB/s]

model.safetensors: 100% 438M/438M [00:02<00:00, 268MB/s]

vocab.txt: 232k/? [00:00<00:00, 13.5MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 4.64kB/s]

FinBERT loaded successfully.
```

```
# Run FinBERT on 1,000 headlines

headlines = df_exp2["headline"].tolist()

print("Running FinBERT on 1,000 headlines...")

finbert_results = finbert(headlines, batch_size=32)
finbert_preds = [r["label"].lower() for r in finbert_results]

df_exp2["finbert_sentiment"] = finbert_preds

print(f"Done. Total predictions: {len(finbert_preds)}")
print(f"\nSentiment distribution:")
print(pd.Series(finbert_preds).value_counts())
```

```
Running FinBERT on 1,000 headlines...
Done. Total predictions: 1000

Sentiment distribution:
positive    514
negative    486
Name: count, dtype: int64
```

```
# Map sentiment to directional prediction and evaluate
# Positive → 1 (price goes up)
# Negative → 0 (price goes down)

df_exp2["finbert_direction"] = df_exp2["finbert_sentiment"].map({
    "positive": 1,
    "negative": 0
})

true_movement = df_exp2["movement"].tolist()
pred_movement = df_exp2["finbert_direction"].tolist()

# Evaluate
acc = accuracy_score(true_movement, pred_movement)
prec = precision_score(true_movement, pred_movement, average="macro")
rec = recall_score(true_movement, pred_movement, average="macro")
f1 = f1_score(true_movement, pred_movement, average="macro")

print("=" * 50)
print("FinBERT – Experiment 2 Results (next-day movement)")
print("=" * 50)
print(f" Directional Accuracy : {acc:.4f} ({acc*100:.2f}%)")
print(f" Precision : {prec:.4f}")
print(f" Recall : {rec:.4f}")
print(f" F1-Score : {f1:.4f}")
print("=" * 50)
```

```
print("\nDetailed Classification Report:")
print(classification_report(true_movement, pred_movement,
    target_names=["Down (0)", "Up (1)"]))
```

```
=====
FinBERT – Experiment 2 Results (next-day movement)
=====
```

```
Directional Accuracy : 0.4880 (48.80%)
Precision             : 0.4878
Recall                : 0.4878
F1-Score              : 0.4878
=====
```

```
Detailed Classification Report:
```

	precision	recall	f1-score	support
Down (0)	0.48	0.47	0.48	492
Up (1)	0.50	0.50	0.50	508
accuracy			0.49	1000
macro avg	0.49	0.49	0.49	1000
weighted avg	0.49	0.49	0.49	1000

```
# Save FinBERT Experiment 2 results
```

```
finbert_exp2_scores = {
    "model": "FinBERT",
    "directional_accuracy": round(acc, 4),
    "precision": round(prec, 4),
    "recall": round(rec, 4),
    "f1_score": round(f1, 4)
}
```

```
df_exp2["finbert_direction"] = pred_movement
```

```
print("FinBERT – Experiment 2 Results Summary")
print(pd.DataFrame([finbert_exp2_scores]))
```

```
# Save to Google Drive
```

```
from google.colab import drive
drive.mount("/content/drive")
```

```
df_exp2.to_csv("/content/drive/MyDrive/Thesis_Data/exp2_finbert_preds.csv", index=False)
print("\nSaved to Google Drive.")
```

```
FinBERT – Experiment 2 Results Summary
```

	model	directional_accuracy	precision	recall	f1_score
0	FinBERT	0.488	0.4878	0.4878	0.4878

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force
```

```
 Saved to Google Drive.
```